compatible with human intuition—*look* like a different kind of machine: a virtual machine, specifically designed to mesh with the human brain and the human hand. The virtual machine known as the Macintosh User Interface is recognizably a machine. It has buttons to press, and slide controls like a hi-fi set. But it is a *virtual* machine. The buttons and sliders are not made of metal or plastic. They are pictures on the screen, and you press them or slide them by moving a virtual finger about the screen. As a human you feel in control, because you are accustomed to moving things around with your finger. I have been an intensive programmer and user of a wide variety of digital computers for twenty-five years, and I can testify that using the Macintosh (or its imitators) is a qualitatively different experience from using any earlier type of computer. There is an effortless, natural feel to it, almost as if the virtual machine were an extension of one's own body. To a remarkable extent the virtual machine allows you to use intuition instead of looking up the manual.

I now turn to the other background idea that we need to import from computer science, the idea of serial and parallel processors. Today's digital computers are mostly serial processors. They have one central calculating mill, a single electronic bottleneck through which all data have to pass when being manipulated. They can create an illusion of doing many things simultaneously because they are so fast. A serial computer is like a chess master 'simultaneously' playing twenty opponents but actually rotating around them. Unlike the chess master, the computer rotates so swiftly and quietly around its tasks that each human user has the illusion of enjoying the computer's exclusive attention. Fundamentally, however, the computer is attending to its users serially.

Recently, as part of the quest for ever more dizzying speeds of performance, engineers have made genuinely parallel-processing machines. One such is the Edinburgh Supercomputer, which I was recently privileged to visit. It consists of a parallel array of some hundreds of 'transputers', each one equivalent in power to a contemporary desk-top computer. The supercomputer works by taking the problem it has been set, subdividing it into smaller tasks that can be tackled independently, and farming out the tasks to gangs of transputers. The transputers take the sub-problem away, solve it, hand in the answer and report for a new task. Meanwhile other gangs of transputers are reporting in with their solutions, so the whole supercomputer gets to the final answer orders of magnitude faster than a normal serial computer could.

I said that an ordinary serial computer can create an illusion of being a parallel processor, by rotating its 'attention' sufficiently fast around a number of tasks. We could say that there is a *virtual* parallel processor sitting atop serial hardware. Dennett's idea is that the human brain has done exactly the reverse. The hardware of the brain is fundamentally parallel, like that of the Edinburgh machine. And it runs software designed to create an illusion of serial processing: a serially processing virtual machine riding on top of parallel architecture. The salient feature of the subjective experience of thinking, Dennett thinks, is the serial 'one-thing-after-another', 'Joycean', stream of consciousness. He believes that most animals lack this serial experience, and use brains directly in their native, parallel-processing mode. Doubtless the human brain, too, uses its parallel architecture directly for many of the routine tasks of keeping a complicated survival machine ticking over. But, in addition, the human brain evolved a software virtual machine to simulate the illusion of a serial processor. The mind, with its serial stream of consciousness, is a virtual machine, a 'user-friendly' way of experiencing the brain, just as the 'Macintosh User Interface' is a user-friendly way of experiencing the physical computer inside its grey box.

It is not obvious why we humans needed a serial virtual machine, when other species seem quite happy with their unadorned parallel machines. Perhaps there is something fundamentally serial about the more difficult tasks that a wild human is called upon to do, or perhaps Dennett is wrong to single us out. He further believes that the development of the serial software has been a largely cultural phenomenon, and again it is not obvious to me why this should be particularly likely. But I should add that, at the time of my writing, Dennett's paper is unpublished and my account is based on recollections of his 1988 Jacobsen Lecture in London. The reader is advised